CSE370:Database Systems

Lecture 6: Normalization For Relational Databases

Database Normalization

Database normalization is the process of decomposing (breaking down) inefficient relations (tables) into smaller relations in order to minimize data redundancy and avoid update anomalies.

Data Redundancy Causes

- Storage Wastage
- Update Anomalies
  - →Insertion Anomaly
  - →Deletion Anomaly
  - →Modification Anomaly

INSERTION ANOMALY

Some data cannot be inserted at all, even though it should be inserted according to business rules.

A new employee cannot be inserted in the EMP_PROJ table unless a project is assigned to the employee, even though it is not the expected insertion rule of the company

EMP_PROJ

			Redundancy	Redundancy	
Ssn	Pnumber	Hours	Ename	Pname	Plocation
123456789	1	32.5	Smith, John B.	ProductX	Bellaire
123456789	2	7.5	Smith, John B.	ProductY	Sugarland
666884444	3	40.0	Narayan, Ramesh K.	ProductZ	Houston
453453453	1	20.0	English, Joyce A.	ProductX	Bellaire
453453453	2	20.0	English, Joyce A.	ProductY	Sugarland
333445555	2	10.0	Wong, Franklin T.	ProductY	Sugarland
333445555	3	10.0	Wong, Franklin T.	ProductZ	Houston
333445555	10	10.0	Wong, Franklin T.	Computerization	Stafford
333445555	20	10.0	Wong, Franklin T.	Reorganization	Houston
999887777	30	30.0	Zelaya, Alicia J.	Newbenefits	Stafford
999887777	10	10.0	Zelaya, Alicia J.	Computerization	Stafford
987987987	10	35.0	Jabbar, Ahmad V.	Computerization	Stafford
987987987	30	5.0	Jabbar, Ahmad V.	Newbenefits	Stafford
987654321	30	20.0	Wallace, Jennifer S.	Newbenefits	Stafford
987654321	20	15.0	Wallace, Jennifer S.	Reorganization	Houston
888665555	20	Null	Borg, James E.	Reorganization	Houston

DELETION ANOMALY

Some data is unintentionally lost due to the deletion of some other data.

The employee named “Narayan” will also be deleted if the project with pnumber 3 is deleted since that employee is assigned only to one project, even though the intention was only to delete the project.

EMP_PROJ

			Redundancy	Redundancy	
Ssn	Pnumber	Hours	Ename	Pname	Plocation
123456789	1	32.5	Smith, John B.	ProductX	Bellaire
123456789	2	7.5	Smith, John B.	ProductY	Sugarland
666884444	3	40.0	Narayan, Ramesh K.	ProductZ	Houston
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MODIFICATION ANOMALY

Same data is stored on multiple rows redundantly, incorrect update may result in data inconsistencies.

Plocation of “ProductX” is updated to “Sugarland” from “Bellaire” for the first row, SSN=’123456789' and Pnumber=1. However, in other rows PLocation for “ProductX” remains “Bellaire”. Thus the data becomes inconsistent.

EMP_PROJ

			Redundancy	Redundancy	
Ssn	Pnumber	Hours	Ename	Pname	Plocation
123456789	1	32.5	Smith, John B.	ProductX	Bellaire
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Functional Dependencies (FDs)

FDs are constraints that are derived from the meaning and interrelationships of the attributes of a table.

X	Y
1	a
2	b
1	a
3	b

- Formally FDs are shown using the notation X → Y
- It means, a set of attributes X functionally determines a set of attributes Y (i.e. Y is functionally dependent on X) such that whenever two rows have the same value for X, the corresponding value for Y must also be the same.
  - If all values of X are unique, then X → Y always holds. If X values are repeated, then X → Y is only true if for each repeated value of X, the corresponding value of Y is the same.

Examples of FDs

- **ID → {Name, CGPA}**
  The student's ID determines the students's name and cgpa in the Student table

- **{ID, CCode} → Grade**
  The Grade is dependent on the ID of the Student and the course code in the 'Grades' table

  - FDs are derived from real-world constraints on the attributes. The primary key of a table will functionally determine all other attributes of the table.

  - A non-prime attribute (attribute not part of pk) may also be functionally dependent on other non-prime attributes.

Student

ID	Name	CGPA
1	Luffy	4.0
2	Sanji	3.5
3	Zoro	3.8

Grades

ID	Ccode	Grade
1	CSE370	A
2	CSE370	B+

Normal Forms

Normal form of a relation(table) indicates the degree to which the relation has been normalized.

- The normal forms are **1NF(first normal form)**, **2NF(second normal form)**, **3NF(third normal form)**, **BCNF(Boyce-Codd normal form)**, **4NF(fourth normal form)** and **5NF(fifth normal form)**.

- Each normal form has some conditions using the keys and FDs of a relation(table). These conditions are used to certify whether that relation is in a particular normal form.

- In practice, schemas are typically normalized up to 3NF (and sometimes BCNF). Higher normal forms, will increase the number of tables, thus will require more joins and negatively impact query performance. Therefore, over-normalization is not practiced in real database systems.

First Normal Form (1NF)

1NF is considered to be part of the definition of a relation

DISALLOWS

- Composite Attributes
- Multivalued Attributes
- Nested Relations (tables); attributes whose values for an individual tuple are non-atomic

Normalize to 1NF

The table below is not in 1NF form as DLocations is multivalued

Department

DName	Dnumber	DmgrID	DLocations
Research	5	E1	{Dhaka, Sylhet, Khulna}
Admin	4	E2	{Dhaka}
HQ	1	E3	{Chittagong}

Department

DName	Dnumber	DmgrID

Dept_Loc

Dnumber	DLocations

A separate table has been created for the multivalued attribute

Second Normal Form (2NF)

Prime attribute: an attribute which is part of the pk
Non-prime attribute: an attribute which is not part of the primary key

TO BE IN 2NF, A RELATION:

- Must also be in 1NF
- Cannot have any partial functional dependency on primary key, thus all non-prime attributes must be fully functionally dependent on the primary key

StudentID	CourseCode	StudentName	Grade

- The primary key of the above table is {StudentID, CourseCode}
- {StudentID, CCode} → Grade. Since Grade depends on the full primary key, it is a full functional dependency
- StudentID → StudentName. Student's name depends only on their ID. Since, a non-prime attribute depends on a part of the primary key, it is a partial functional dependency

The table below is not in 2NF form as partial dependency exists on primary key due to FD2 and FD3.

Emp_Proj

EID	Pno	Hours	Ename	Pname	Plocation

Functional Dependencies:

FD1: EID, Pno → Grades
FD2: EID → Ename
FD3: Pno → Pname, Plocation

Normalize to 2NF

Employee

EID	Ename

Project

Pno	Pname	Plocation

Emp_Proj

EID	Pno	Hours

Note: If functional dependency of any non-prime attribute is not mentioned, then assume full dependency on primary key.

Third Normal Form (3NF)

If X → Y and Y → Z, where X is pk and Y is non-prime attribute, then X → Z is a transitive dependency

TO BE IN 3NF, A RELATION:

- Must also be in 2NF
- Cannot have any transitive functional dependency, thus no non-prime attribute should be transitively dependent on the primary key

<u>StudentID</u>	StudentName	DeptID	DName

- The primary key of the above table is StudentID
- StudentID → DeptID and DeptID → DName, therefore, StudentID → DName is a transitive dependency
- StudentID → StudentName is a non-transitive dependency

Normalize to 3NF

The table below is not in 3NF form as transitive dependency exists on primary key due to FD2.

Emp_Dept

EID	Ename	DOB	Address	Dno	Dname	DMgr

Functional Dependencies:

FD1: EID → Ename, DOB, Address, Dno
FD2: DNo → DName, DMgr

Employee

EID	Ename	DOB	Address	Dno

Department

Dno	Dname	DMgr

Note: If any non-prime attribute is functionally dependent on any other non-prime attribute, it creates transitive dependency on the primary key

Normalization Example

Student

StID	Course Code	Semester	Student Name	Attempted Credits	Course Title	Grade	Proj ID	Project Marks	Project Name

Additional Functional Dependencies

FD1: StudentID → StudentName
FD2: CourseCode → CourseTitle
FD3: StudentID, Semester → AttemptedCredits
FD4: ProjID → ProjectName, ProjectMarks

Question: Identify and explain which normal form the above schema is in. Step by step normalize it until 3rd normal form. You MUST show the complete "current" schema after each normalization step.

Normalization Example - Solution (1)

The given schema is in first normal form (1NF) because there are no multivalued/composite attributes and there are no nested tables. It is not in second normal form (2NF) because partial functional dependency on primary key exists due to FD1, FD2 and FD3.

2NF Normalization

Student

StID	StudentName

Course

CourseCode	Course Title

Student_Credits

StID	Semester	Attempted Credits

Student_Grades

StID	CourseCode	Semester	Grade	ProjID	Project Marks	Project Name

Normalization Example - Solution (2)

The current schema is in 2NF but not in third normal form (3NF) yet, because transitive functional dependency on primary key exists due to FD4 in Student_Grades table.

3NF Normalization

Student

StID	StudentName

Course

CourseCode	Course Title

Student_Credits

StID	Semester	Attempted Credits

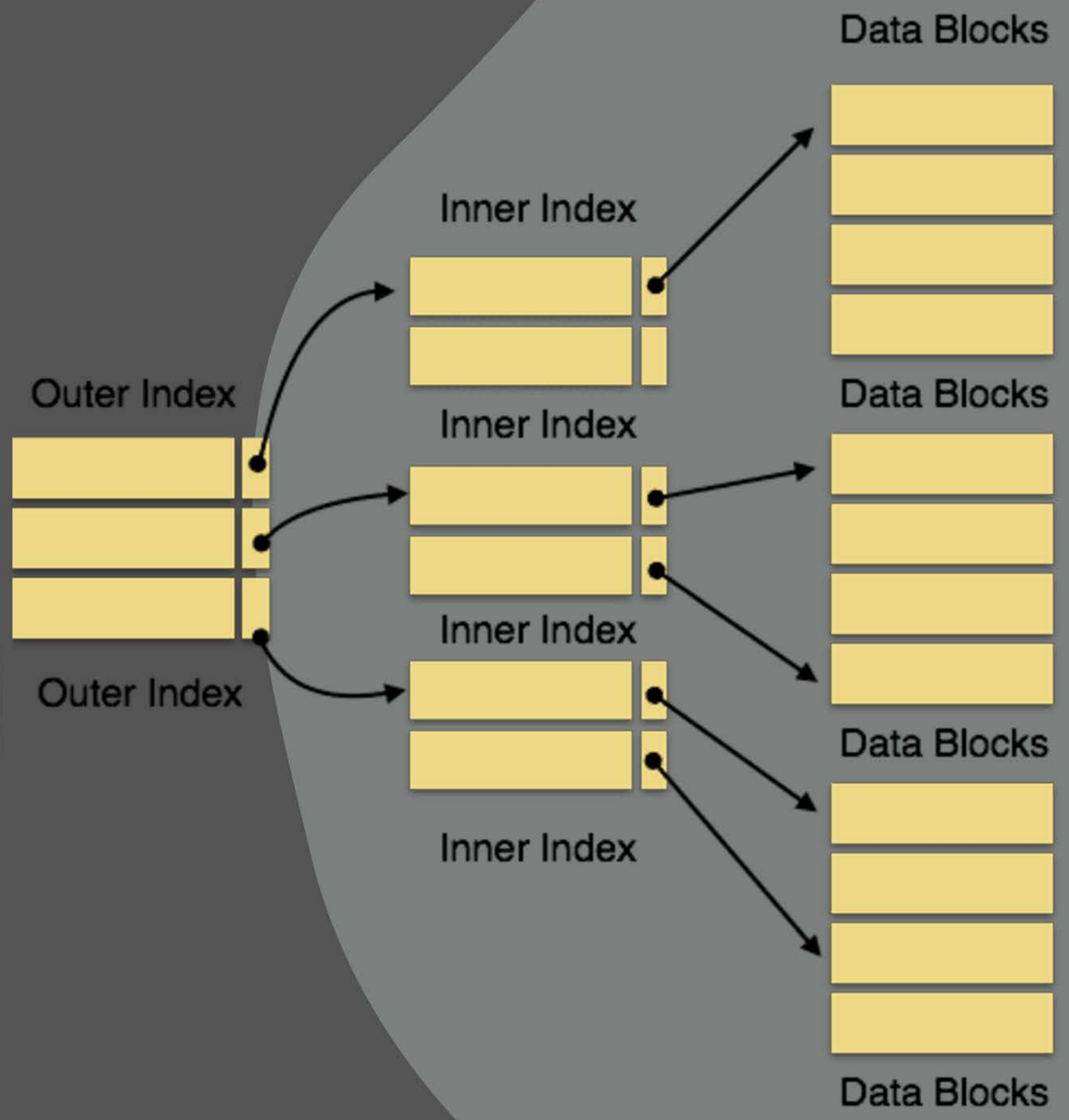
Student_Grades

StID	Course Code	Semester	Grade	ProjID

Student_Projects

ProjID	Project Marks	Project Name

What Next?

LECTURE 7: INDEXING

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